EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Introduction; Vapour Pressure; Henry's Law

1. The value of Henry's law constant for some gases at 293 K is given below. Arrange the gases in the increasing order of their solubility.

He : 144.97 kbar, H_2 : 69.16 kbar, N_2 : 76.48 kbar, O_2 : 34.86 kbar

(A) He < N_2 < H_2 < O_2 (B) O_2 < H_2 < N_2 < He (C) H_2 < N_2 < O_2 < He (D) He < O_2 < N_2 < H_2

CLQ088

2. A gas 'X' is dissolved in water at 2 bar pressure. Its mole fraction is 0.02 in solution. The mole fraction of water when the pressure of gas is doubled at the same temperature.

(A) 0.04 (B) 0.98 (C) 0.96 (D) 0.02

CLQ093

3. If a ground water contains H_2S at concentration of 2 mg/l, determine the pressure of H_2S in head space of a closed tank containing the ground water at 20°C. Given that for H_2S, Henry's constant is equal to 6.8×10^3 bar at 20°C.

(A) 720 Pa (B) 77×10^2 Pa (C) 553 Pa (D) 55×10^2 Pa

CLQ164

4. A pressure cooker reduces cooking time for food because -

(A) The higher pressure inside the cooker crushes the food material

(B) Cooking involves chemical changes helped by a rise in temperature

(C) Heat is more evenly distributed in the cooking space

(D) Boiling point of water involved in cooking is increased

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CLQ165

5. An unknown solution [Molecular wt of solute = 250] is 20% $\frac{w}{W}$. Molarity of solution is ..

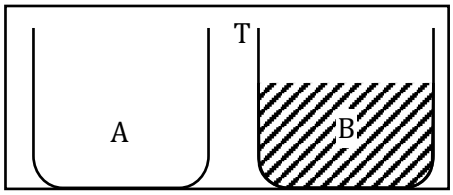
[Given density of solution = 1.25 g/mL]

(A) 1 M (B) 2 M (C) 3 M (D) 4 M

CLQ086

Raoult's Law; Ideal Solutions

6. Two beakers A and B containing pure water and aqueous solution are shown. What will be observed gradually with passage of time?



(A) Level of water in beaker A becomes lower than that in beaker B

(B) Level of water in beakers A and B remains constant

(C) Level of water in beaker A increases and that in beaker B decreases

(D) Level of both the beakers increases

CLQ089

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7. The vapour pressure ratio P_A^o : P_B^o for the two volatile liquid A and B are 1:2 and mole ratio is $X_A : X_B = 1: 2$. What is the mole fraction of A in vapour phase ?

(A) 0.33 (B) 0.25 (C) 0.20 (D) 0.52

CLQ094

8. Liquids A and B form an ideal solution. A certain solution of A and B contains 25 mole percent of A, whereas the vapours in the equilibrium with the solution at 298 K contains 50 mole percent of A. The ratio of vapour pressures of pure A to that of pure B at 298 K, is

(A) 1 : 1 (B) 3 : 1 (C) 1 : 3 (D) 2 : 1

CLQ103

9. Two components A and B form an ideal solution. The mole fractions of A and B in ideal solution are X_A and X_B while that in vapour phase, these components have their mole fractions as Y_A and Y_B. Then, the slope and intercept of plot of $\frac{1}{Y_A}$ vs. $\frac{1}{X_A}$ will be :-

(A) $\frac{P_B^o}{P_A^o}, \frac{P_A^o - P_B^o}{P_A^o}$ (B) $\frac{P_A^o}{P_B^o}, \frac{P_B^o - P_A^o}{P_B^o}$ (C) $\frac{P_B^o}{P_A^o}, \frac{P_B^o}{P_B^o - P_A^o}$ (D) $P_A^o - P_B^o, \frac{P_A^o}{P_B^o}$

CLQ133

10. The boiling point of C_6H_6, CH_3OH, $C_6H_5NH_2$ and $C_6H_5NO_2$ are 80°C, 65°C, 184°C and 212°C respectively which will show highest vapour pressure at room temperature :

(A) C_6H_6 (B) CH_3OH (C) $C_6H_5NH_2$ (D) $C_6H_5NO_2$

CLQ138

11. 1 mole of heptane (V. P. = 92 mm of Hg) was mixed with 4 moles of octane (V. P. = 31mm of Hg). The vapour pressure of resulting ideal solution is :

(A) 46.2 mm of Hg (B) 40.0 mm of Hg (C) 43.2 mm of Hg (D) 38.4 mm of Hg

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CLQ141

12. Mole fraction of A vapours above solution in mixture of A and B ($X_A = 0.4$) will be :- (P_A^o = 100mm, P_B^o = 200mm)

(A) 0.4 (B) 0.8 (C) 0.25 (D) None

CLQ142

13. The vapour pressure of a solution having solid as solute and liquid as solvent is :

(A) Directly proportional to mole fraction of the solvent

(B) Inversely proportional to mole fraction of the solvent

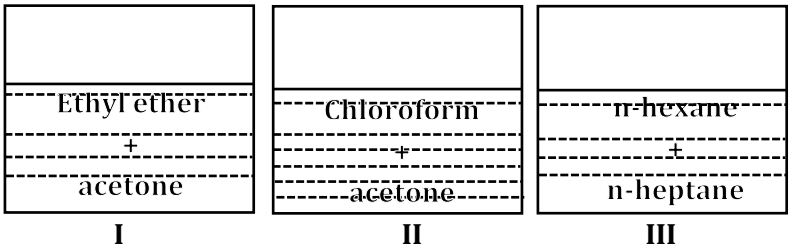
(C) Directly proportional to mole fraction of the solute

(D) Inversely propartional to mole fraction of the solute

CLQ143

Non - Ideal Solutions

14. State T(true)/F(false) for following statements regarding given liquid solutions :



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(a) Final volume of solutions if equal volume of compounds are mixed, is in the order: $V_I = V_{II} < V_{III}$
(b) ΔS_{mix} for solution I and solution II is positive but for solution III it is zero.
(c) Enthalpy change of mixing is negative for solution II and solution III but positive for solution I.
(A) TFF (B) FFF (C) TTF (D) FTT

CLQ095

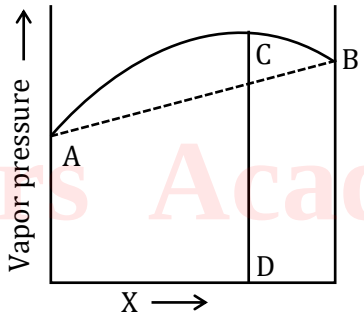
15. The vapour pressure of the solution of two liquids A(P° = 80 mm) and B(P° = 120 mm) is found to be 100 mm when $X_A = 0.4$. The result shows that
(A) Solution exhibits ideal behaviour
(B) Solution shows positive deviation from ideal behaviour
(C) Solution shows negative deviation from ideal behaviour
(D) Attractive intermolecular forces between A–A in pure liquid A and B–B in pure liquid B are stronger than those between A–B when mixed in solution.

CLQ096

16. Which of the following mixture of solutions cannot form any type of azeotropes ?
(A) $HCl + H_2O$ (B) $CH_3COCH_3 + CHCl_3$ (C) $C_2H_5Br + C_2H_5I$ (D) $C_2H_5OH + H_2O$

CLQ097

17. The diagram given below is a vapour pressure-composition diagram for a binary solution of A and B. In the solution, A–B interactions are –



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(A) similar to A–A and B–B interactions (B) greater than A–A and B–B interactions
(C) smaller than A–A and B–B interactions (D) unpredictable

CLQ098

18. A binary solution of liquid A and B will show positive deviation from Raoult's law if it fulfils the following condition :-
(1) $P_A > P_A^o X_A$ and $P_B > P_B^o X_B$
(2) The inter molecular force of A–B < A–A, B–B
(3) ΔH_{mix} is positive
(4) ΔV_{mix} is negative
(A) 1, 2 and 4 are True (B) 1, 2 are wrong
(C) 1, 2, 3 are true (D) 1, 3, 4 are true

CLQ100

19. An azeotropic mixture of two liquid has a boiling point higher than either of them when it:-
(A) show positive deviation from Raoult's law
(B) shows negative deviation from Raoult's law
(C) show ideal behaviour
(D) is saturated

CLQ102

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Colligative Properties

20. A liquid freezes at 7°C and boils at 77°C. If values of K_f and K_b of this liquid are 5.6 K-kg/mole and 2.5 K-kg/mole respectively, then ratio of molar latent heat of fusion (in kJ/mol) to molar latent heat of vaporisation (in kJ/mol) for the liquid is -

(A) 1 : 1 (B) 7 : 2 (C) 2 : 7 (D) 7 : 3

CLQ107

21. A solution containing 0.1 g of a non-voltatile organic substance P(molecular mass 100) in 100g of benzene raises the boiling point of benzene by 0.2°C, while a solution containing 0.1 g of another non-volatile substance Q in the same amount of benzene raises the boiling point of benzene by 0.4°C. What is the ratio of molecular masses of P and Q ?

(A) 1 : 2 (B) 2 : 1 (C) 1 : 4 (D) 4 : 1

CLQ109

22. Elevation of boiling point of 1 molar aqueous glucose solution (density = 1.2 g/ml) is

(A) K_b (B) 1.20 K_b (C) 1.02 K_b (D) 0.98 K_b

CLQ112

23. In three beakers labelled as (A), (B) and (C), 100 mL of water, 100 mL of 1M aqueous solution of glucose and 100 mL of 0.5 M aqueous solution of urea are taken respectively and kept at same temperature.

Which of the following statements is correct ?

(A) Vapour pressure in all the three beakers is same.

(B) Vapour pressure of beaker B is highest among all.

(C) Vapour pressure of beaker C is highest among all.

(D) Vapour pressure of beaker B is lower than that of C and vapour pressure of beaker C is lower than that of A.

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CLQ113

24. In the depression of freezing point experiment, it is found that

(I) The vapour pressure of the solution is less than that of pure solvent.

(II) The vapour pressure of the solution is more than that of pure solvent.

(III) Only solute molecules solidify at the freezing point.

(IV) Only solvent molecules solidify at the freezing point.

(A) I, II (B) II, III (C) I, IV (D) I, II, III

CLQ114

25. An aqueous solution contains 3% and 1.8% by wt. urea and glucose respectively. What is the freezing point of solution ? (K_f = 1.86°C/m):-

(A) –1.172°C (B) –2.27°C (C) –1.5°C (D) None of these

CLQ127

26. The freezing point of a 4% aqueous solution of A' is equal to the freezing point of 10% aqueous solution of B'. If the molecular weight of 'A' is 60 than the molecular weight of B' will be:-

(A) 160 (B) 90 (C) 45 (D) 180

CLQ130

27. The vapour pressure of a solvent decreased by 10 mm of Hg when a non-volatile solute was added to the solvent. The mole fraction of solute in solution is 0.2, what would be mole fraction of the solvent if decrease in vapour pressure is 20 mm of Hg

(A) 0.2 (B) 0.4 (C) 0.6 (D) 0.8

CLQ146

For More Join: @IITJEE_Advanced

28. At a certain temperature when glucose is dissolved in water lowering in vapour pressure is 0.6 mmHg. What will be the vapour pressure of glucose solution if molality is 1/18?

(A) 200 mm Hg (B) 300 mm Hg (C) 600 mm Hg (D) 720 mm Hg

CLQ111

29. What should be the freezing point of aqueous solution containing 17 gm of C_2H_5OH in 1000 gm of water (water K_f = 1.86 °C – kg mol^{-1})

(A) – 0.69°C (B) – 0.34°C (C) 0.0°C (D) 0.34°C

CLQ148

30. The vapour pressure of a saturated solution of sparingly soluble salt (XCl_3) was 17.20 mm Hg at 27°C. If the vapour pressure of pure H_2O is 17.25 mm Hg at 300 K, what is the solubility of sparingly soluble salt XCl_3 in mole/Litre.

(A) 4.04×10^{-2} (B) 8.08×10^{-2} (C) 2.02×10^{-2} (D) 4.04×10^{-3}

CLQ149

31. **Statement-1** : Addition of ethylene glycol (non-volatile) to water lowers the freezing point of water hence used as antifreeze.

Statement-2 : Addition of any substance to water lowers its freezing's point of water.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

CLQ150

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Abnormal Colligative Properties

32. Consider following solutions:

(P) 1M aqueous glucose solution

(Q) 1M aqueous sodium chloride solution

(R) 1M aqueous ammonium phosphate solution

(S) 1M benzoic acid in benzene (α = 1)

Select the INCORRECT statements -

(A) $\pi_P = \pi_S$ (B) $\pi_R > \pi_S$ (C) $\pi_R > \pi_Q$ (D) $\pi_Q > \pi_P$

CLQ115

33. When 20 g of naphtholic acid ($C_{11}H_8O_2$) is dissolved in 50 g of benzene (K_f = 1.72 K kg mol^{-1}), a freezing point depression of 2 K is observed. The Van't Hoff Factor (i) is

(A) 0.5 (B) 1 (C) 2 (D) 3

CLQ117

34. Which of the following 0.1 M aqueous solutions will have lowest freezing point ?

(A) Potassium sulphate (B) Sodium chloride

(C) Urea (D) Glucose

CLQ119

35. Amongst the following, the solution which shows highest osmotic pressure is

(A) 0.05 M NaCl (B) 0.10 M $BaCl_2$ (C) 0.05 M $FeCl_3$ (D) 0.05 M Na_2SO_4

CLQ122

For More Join: @IITJEE_Advanced

36. Which of the following is isotonic with 15% $\frac{w}{w}$ glucose solution (sp. gr = 1.2) :

(A) 1 M $CaCl_2$ (B) 1 N $CaCl_2$ (C) 0.5 M NaCl (D) 0.5 M urea

CLQ123

37. Select the incorrect option.

(A) For solution showing +ve deviation, there is minima in boiling point curve

(B) Boiling Point of aqueous solution of 0.1 M, KCl is greater than that of 0.1 M glucose solution

(C) Osmotic pressure is a colligative property

(D) Colligative properties are defined for non-ideal solutions

CLQ125

38. In a solution, if three molecules of solute undergoes association as : $3A \rightleftharpoons A_3$, then to evaluate degree of association 'a' the formula will be :-

(A) $\alpha = \frac{3(1-i)}{2}$ (B) $\alpha = \frac{i-1}{2}$ (C) $\alpha = \frac{3(i-1)}{2}$ (D) $\alpha = \frac{i-1}{3}$

CLQ126

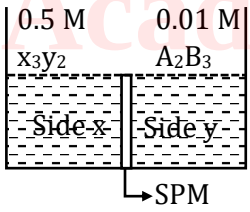
39. A certain substance 'A' tetramerises in water to the extent of 80%. A solution of 2.5 g of A in 100 g of water lowers the freezing point by 0.3°C. The molar mass of A is :-

(A) 122 (B) 31 (C) 244 (D) 62

CLQ129

40. X_3Y_2 (i = 5) when reacted with A_2B_3 (i = 5) in aqueous solution given brown colour. These are separated by a semipermeable membrane AB as shown. Due to osmosis there is :-

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(A) Brown colour formation in side x (B) Brown colour formation in side y

(C) Formation in both of sides x and y (D) No brown colour formation

CLQ131

41. If mole fraction of the solvent in solution decreases then :

(A) Vapour pressure of solution increases (B) B. P. decreases

(C) Osmotic pressure increases (D) All are correct

CLQ151

42. 5% (w/v) solution of sucrose is isotonic with 1% (w/v) solution of a compound 'A' then the molecular weight of compound 'A' is -

(A) 32.4 (B) 68.4 (C) 121.6 (D) 34.2

CLQ152

43. Osmotic pressure of a sugar solution at 24°C is 2.5 atmosphere. The concentration of the solution in mole per litre is :

(A) 10.25 (B) 1.025 (C) 1025 (D) 0.1025

CLQ153

For More Join: @IITJEE_Advanced

44. If a 6.84% (wt./vol.) solution of cane-sugar (mol. wt. 342) is isotonic with 1.52% (wt./vol.) solution of thiocarbamide, then the molecular weight of thiocarbamide is :

(A) 152 (B) 76 (C) 60 (D) 180

CLQ155

45. Which of the following aqueous solution will show maximum vapour pressure at 300 K?

(A) 1 M NaCl (B) 1 M $CaCl_2$ (C) 1 M $AlCl_3$ (D) 1 M $C_{12}H_{22}O_{11}$

CLQ156

46. 1.0 molal aqueous solution of an electrolyte A_2B_3 is 60% ionised. The boiling point of the solution at 1 atm is ($K_{b(H_2O)} = 0.52\ K\,kg\,mol^{-1}$)

(A) 274.76 K (B) 377 K (C) 376.4 K (D) 374.76 K

CLQ157

47. The freezing point depression of a 0.1 M a solution of weak acid (HX) is –0.20°C.
What is the value of equilibrium constant for the reaction?
$HX\ (aq) \rightleftharpoons H^+(aq) + X^-\ (aq)$
[Given : K_f for water = 1.8 kg mol^{-1} K. & Molality = Molarity]

(A) 1.46×10^{-4} (B) 1.35×10^{-3} (C) 1.21×10^{-2} (D) 1.35×10^{-4}

CLQ158

48. The value of observed and calculated molecular weight of silver nitrate are 92.64 and 170 respectively. The degree of dissociation of silver nitrate is :

(A) 60% (B) 83.5 % (C) 46.7% (D) 60.23%

CLQ161

49. What is the freezing point of a solution containing 8.1 gm. of HBr in 100gm. water assuming the acid to be 90% ionised (K_f for water=1.86 K molality^{-1}) :-

(A) 0.85°C (B) –3.53°C (C) 0°C (D) –0.35°C

CLQ162

50. Assuming each salt to be 90 % dissociated, which of the following will have highest boiling point?

(A) Decimolar $Al_2(SO_4)_3$
(B) Decimolar $BaCl_2$
(C) Decimolar Na_2SO_4
(D) A solution obtained by mixing equal volumes of (B) and (C)

CLQ163

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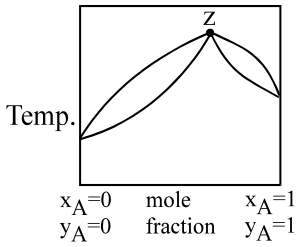
For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

Non - Ideal Solutions

1. A liquid mixture having composition corresponding to point z in the figure shown is subjected to distillation at constant pressure.



Which of the following statement is correct about the process?

(A) The composition of distillate differs from the mixture

(B) The boiling point goes on changing

(C) The mixture has lowest vapour pressure than for any other composition.

(D) Composition of an azeotrope alters on changing the external pressure.

CLQ167

2. Which of the following is correct for a non-ideal solution of liquids A and B, showing negative deviation?

(A) ΔH_{mix} = –ve (B) ΔV_{mix} = –ve (C) ΔS_{mix} = +ve (D) ΔG_{mix} = –ve

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CLQ168

Abnormal Colligative Properties

3. Which one of the following statements is True?

(A) Raoult's law states that the vapour pressure of a component over a solution is proportional to its mole fraction.

(B) The osmotic pressure (π) of a dilute solution is given by the equation π = iMRT where M is the molarity of the solution.

(C) The correct order of osmotic pressure for 0.01M aqueous solution of each compound is $BaCl_2$ > KCl > CH_3COOH > Sucrose

(D) Two sucrose solutions of same molality prepared in different solvents will have the same freezing point depression

CLQ134

COMPREHENSION TYPE QUESTIONS

Colligative Properties

Paragraph for Question No. 4 to 6

An ideal solution is obtained by mixing a non-volatile solute B with a volatile solvent A (molar mass = 60). If the mass ratio of A : B in solution is 10 : 1 and vapour pressure of pure A is 400 mm and vapour pressure decreases by 4% on forming the above solution at 300K.

4. The mole fraction of solute in the solution is -

(A) 0.96 (B) 0.04 (C) 0.16 (D) 0.84

CLQ170

For More Join: @IITJEE_Advanced

5. The molality the solution is -

(A) 1 (B) $\frac{36}{25}$ (C) $\frac{25}{36}$ (D) $\frac{36}{25}$

CLQ170

6. The molar mass of B in the solution is -

(A) 1440 (B) 14400 (C) 4 (D) 144

CLQ170

MATRIX MATCH TYPE QUESTION

Non - Ideal Solutions

Table type question :

Column-I	Column-II	Column-III
(1) $C_6H_6 + C_6H_5–CH_3$	(a) $\Delta H = 0$	(P) $\Delta G = -ve$
(2) $CHCl_3 + CH_3COCH_3$	(b) $\Delta H = +ve$	(Q) Form minimum boiling azerotrope
(3) $CCl_4 + CH_3COCH_3$	(c) $\Delta H = -ve$	(R) Form maximum boiling azerotrope
(4) $C_2H_5OH + H_2O$	(d) $\Delta S = +ve$	(S) No azeotrope

7. Select the correct match -

(A) 1, a, R (B) 2, b, P (C) 3, b, Q (D) 4, d, R

CLQ171

8. Select the correct match -

(A) 1, d, S (B) 3, b, S (C) 2, b, S (D) 4, c, P

CLQ171

9. Select the incorrect match -

(A) 1, d, P (B) 2, c, R (C) 3, d, S (D) 3, b, P

CLQ171

Abnormal Colligative Properties

10. **Match the column :**

Column-I
(Colligative properties)
(Assume m = M)

(A) $\Delta T_f = 0.3 \times K_f$

(B) $\Delta T_b = 0.28 \times K_b$

(C) $\pi = 0.19 \times RT$

(D) $\frac{P^0 - P}{P^0} = \frac{\left(\frac{\Delta T_f}{K_f}\right)}{\left(\frac{1000}{18}\right) + \left(\frac{\Delta T_f}{K_f}\right)}$

Column-II
(Aqueous solution)

(P) 0.1 m – $Ca(NO_3)_2$

(Q) 0.14 m – NaBr

(R) 0.1 m – $MgCl_2(\alpha = 0.9)$

(S) 0.28 m – Urea

(T) 0.1 m – HA (monobasic acid, $K_a = 0.81$)

CLQ172

For More Join: @IITJEE_Advanced

EXERCISE – S

Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

Introduction, Vapour Pressure, Henry's Law

1. The Henry law constant for dissolution of a gas in aqueous medium is 3×10^2 atm. At what partial pressure of the gas (in atm), the molality of gas in aqueous solution will $\frac{5}{9}$ m. be

CLQ209

Raoult's Law; Ideal Solutions

2. The vapour pressure of ethanol and methanol are 44.0 mm and 88.0 mm Hg respectively. An ideal solution is prepared at the same temperature by mixing 69 g of ethanol with 40 g of methanol. Calculate total vapour pressure of the solution. (in mm Hg)

CLQ174

Colligative Properties

3. The vapour pressure of an aqueous solution is found to be 750 torr at certain temperature 'T'. If 'T' is the temperature at which pure water boils under atmospheric pressure, calculate the value of atmospheric pressure (in Torr). The boiling point of solution is 101.04°C. (K_b = 0.52 K kg mol^{-1}).

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CLQ191

4. The mass of a non-volatile, non-electrolyte solute (in grams) (molar mass = 50 g mol^{-1}) needed to be dissolved in 114 g octane to reduce its vapour pressure by 75%, is :-

CLQ183

5. The freezing point of a solution containing 2.40 g of a compound in 60.0 g of benzene is 0.10°C lower than that of pure benzene. What is the molecular weight of the compound (in amu)? (K_f is 5.12°C/m for benzene)

CLQ188

6. An aqueous solution containing 288 gm of a non-volatile compound having the stoichiometric composition $C_xH_{2x}O_x$ in 90 gm water boils at 101.38°C at 1.00 atmospheric pressure. Find value of 'x' ? $K_b(H_2O)$ = 0.52 K mol^{-1} kg

CLQ201

Abnormal Colligative Properties

7. **2.0** g of benzoic acid dissolved in 25.0g of benzene shows a depression in freezing point equal to 1.96K. Molal depression constant (K_f) of benzene is 4.9 K.kg.mol^{-1}. What is the percentage association of the acid?

CLQ197

8. Cane sugar undergoes the inversion as follows

$$C_{12}H_{22}O_{11} + H_2O \rightarrow C_6H_{12}O_6 + C_6H_{12}O_6$$

If solution of 0.025 moles of sugar in 200 gm of water show depression in freezing point 0.372°C, then what % sucrose has inverted? ($K_f(H_2O)$ = 1.86 K kg mol^{-1})

CLQ205

For More Join: @IITJEE_Advanced

Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

Introduction, Vapour Pressure, Henry's Law

9. How many grams O_2 gas will dissolve in 100 gm water at 9 bar and 27°C? (K_H = 40 Kbar)

CLQ208

10. The solubility of N_2 in water at 300 K and 500 torr partial pressure is 0.01 g L^{-1}. If solubility of N_2 (in g L^{-1}) at 750 torr partial pressure is 's' then find the value of 100s :

CLQ211

Raoult's Law; Ideal Solutions

11. At 25°C, the vapour pressure of methyl alcohol is 96.0 torr. What is the mole fraction of CH_3OH in a solution in which the (partial) vapour pressure of CH_3OH is 24.0 torr at 25°C?

CLQ173

12. Liquids 'A" and 'B' form an ideal solution. Calculate the mole-fraction of 'A' in the vapours above the liquid solution containing the liquids 'A' and 'B' in 2 : 3 mole ratio, at equilibrium.
[Given : P_A^o = 0.4 atm, P_B^o = 0.8 atm]

CLQ176

13. Liquids 'P' and 'Q' form an ideal solution. At equilibrium, the vapours contain 40% molecules of 'P'. Calculate the vapour pressure of solution. (in Bar)
[Given : P_P^o = 0.4 bar, P_Q^o = 0.6 bar]

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CLQ177

14. Find number of mixtures among following for which enthalpy change of mixture is zero.
benzene + toluene, n-hexane + n-heptane, ethyl bromide + ethyl iodide, $CCl_4 + SiCl_4$, $CCl_4 + CHCl_3$, acetone + benzene, $C_2H_4Br_2 + C_2H_4Cl_2$, $C_2H_5Br + C_2H_5Cl$, cyclohexane + ethanol

CLQ206

15. The molar volume of liquid benzene (density = 0.877 g ml^{-1}) increases by a factor of 2750 as it vaporizes at 20°C and that of liquid toluene (density = 0.867gml^{-1}) increases by a factor of 7720 at 20°C. A Solution of benzene & toluene has a vapour pressure of 46.0 torr. Find the mole fraction of benzene in the vapour above the solution.

CLQ212

Non - Ideal Solutions

16. Determine the number of mixtures which show negative deviation from ideal behaviour among following.
Acetone + aniline, Acetone + ethanol, Chloroform + Dry ethyl ether, $H_2O + HNO_3$, $CCl_4 + CH_3OH$, Cyclohexane + Ethanol, Acetic acid + Pyridine, Benzene + Chloroform.

CLQ207

Colligative Properties

17. Calculate the relative lowering in vapour pressure if 100 g of a nonvolatile solute (mol.wt.100) are dissolved in 432 g water.

CLQ181

For More Join: @IITJEE_Advanced

18. The vapour pressure of pure benzene at 30° C is 640 mm of Hg and the vapour pressure of a solution of a solute in C_6H_6 at the same temperature is 624 mm of Hg. Calculate molality of solution.

CLQ182

19. The vapour pressure of water is 17.54 mm Hg at 293 K. Calculate vapour pressure of 0.5 molal solution of a solute in it (in mm Hg). [Give your answer as nearest integer]

CLQ184

20. Pure benzene freezes at 5.45°C. A solution containing 6.72 g of $C_2H_2Cl_4$ in 120 g of benzene was observed to freeze at 3.75°C. What is the molal freezing point depression constant of benzene (in K-Kg mol^{-1}) ?

CLQ187

21. Calculate the amount of ice (in grams) that will separate out by cooling a solution containing 50g of ethylene glycol in 200 g water to –9.3°C. (K_f for water = 1.86 K mol^{-1} kg)

CLQ190

22. Calculate the osmotic pressure (in atm) of 12% (w/v) aq. urea solution at 27°C.

CLQ192

23. Calculate the osmotic pressure (in atm) of a solution containing 18 gm glucose and 17.1 gm cane-sugar ($C_{12}H_{22}O_{11}$) per litre, at 27°C.

CLQ193

24. A solution of crab hemocyanin, a pigmented protein extracted from crabs, was prepared by dissolving 0.750 g in 125 cm^3 of an aqueous medium. At 4°C, an osmotic pressure rise of 2.77 mm of the solution was observed. The solution had a density of 1.013 g/cm^3. Determine the molar mass of the protein. (in g/mol) (g = 10 m/s^2), (R = 0.08 l-atm/K-mol)

CLQ200

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Abnormal Colligative Properties

25. What is the freezing point (in degrees) of a solution containing 8.1 g of HBr in 100g water assuming the acid to be 90% ionised (K_f for water = 1.86 K $molality^{-1}$) [Atomic mass : Br = 80]

CLQ195

26. A 0.2 molal aqueous solution of a weak acid (HX) is 20% ionised. What is the elevation in boiling point of this solution (In Kelvin) ? (given K_b = 0.52°C kg mol^{-1} for H_2O)

CLQ196

27. A decimolar solution of potassium ferrocyanide is 50% dissociated at 300K. Calculate the osmotic pressure of the solution. (in S.I. units) (R=8.314 JK^{-1} mol^{-1})

CLQ198

28. A complex is represented as $CoCl_3 \cdot xNH_3$. It's 0.1 molal solution in a solution shows ΔT_f = 0.558°C. K_f for H_2O is 1.86 K mol^{-1} kg. Assuming 100% ionisation of complex and coordination number of Co is six, calculate the number of chloride ions outside the coordination sphere in the complex ?

CLQ202

29. Phenol associates in benzene to a certain extent to form a dimer. A solution containing 18.8×10^{-3} kg phenol in 1 kg of benzene has its freezing point depressed by 0.768 K. Calculate the fraction of phenol that has dimerised. K_f for benzene = 5.12 kg mol^{-1}K.

CLQ203

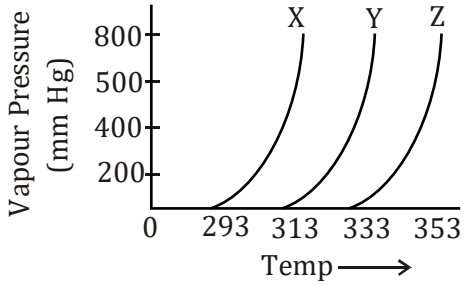
30. The cryoscopic constant for acetic acid is 3.6 K kg/mol. A solution of 1 g of a hydrocarbon (C_XH_Y) in 100 g of acetic acid freezes at 16.14°C instead of the usual 16.60°C. The hydrocarbon contains 92.3% carbon. Find value of (X + Y) ?

CLQ204

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. A graph of vapour pressure and temperature for three different liquids X, Y and Z is shown below :



The following inferences are made :

(A) X has higher intermolecular interactions compared to Y.

(B) X has lower intermolecular interactions compared to Y.

(C) Z has lower intermolecular interactions compared to Y.

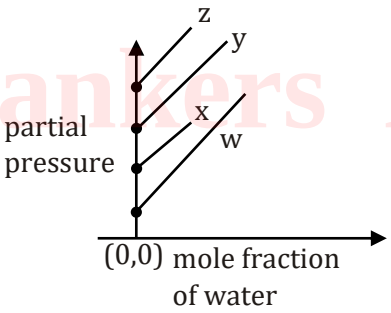
The correct inference(s) is/are : **[JEE (Main) 2020]**

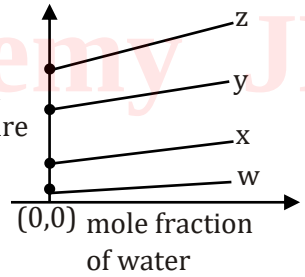
(1) A (2) (C) (3) (B) (4) (A) and (C)

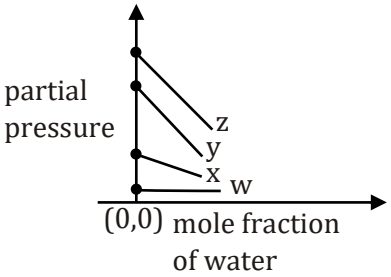
CLQ041

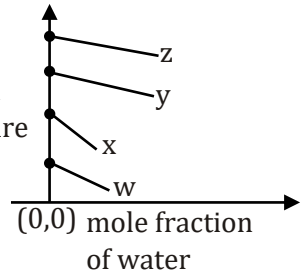
2. For the solution of the gases w, x, y and z in water at 298K, the Henrys law constants (K_H) are 0.5, 2, 35 and 40 kbar, respectively. The correct plot for the given data is :- **[JEE (Main) 2019]**

Rankers Academy JEE

(1)


(2)


(3)


(4)


CLQ034

3. At 35°C, the vapour pressure of CS_2 is 512 mm Hg and that of acetone is 344 mm Hg. A solution of CS_2 in acetone has a total vapour pressure of 600 mm Hg. The false statement amongst the following is : **[JEE (Main) 2020]**

(1) heat must be absorbed in order to produce the solution at 35°C

(2) Raoult's law is not obeyed by this system

(3) a mixture of 100 mL CS_2 and 100 mL acetone has a volume < 200 mL

(4) CS_2 and acetone are less attracted to each other than to themselves

CLQ040

For More Join: @IITJEE_Advanced

4. At 300 K, the vapour pressure of a solution containing 1 mole of n-hexane and 3 moles of n-heptane is 550 mm of Hg. At the same temperature, if one more mole of n-heptane is added to this solution, the vapour pressure of the solution increases by 10 mm of Hg. What is the vapour pressure in mm Hg of n-heptane in its pure state _____ ? **[JEE (Main) 2020]**

CLQ045

5. Liquid 'M' and liquid 'N' form an ideal solution. The vapour pressures of pure liquids 'M' and 'N' are 450 and 700 mmHg, respectively, at the same temperature. Then correct statement is:
(x_M = Mole fraction of 'M' in solution ;
x_N = Mole fraction of 'N' in solution ;
y_M = Mole fraction of 'M' in vapour phase ;
y_N = Mole fraction of 'N' in vapour phase) **[JEE (Main) 2019]**

(1) $(x_M - y_M) < (x_N - y_N)$

(2) $\frac{x_M}{x_N} < \frac{y_M}{y_N}$

(3) $\frac{x_M}{x_N} > \frac{y_M}{y_N}$

(4) $\frac{x_M}{x_N} = \frac{y_M}{y_N}$

CLQ037

6. A solution of two miscible liquids showing negative deviation from Raoult's law will have : **[JEE (Main) 2024]**

(1) increased vapour pressure, increased boiling point
(2) increased vapour pressure, decreased boiling point
(3) decreased vapour pressure, decreased boiling point
(4) decreased vapour pressure, increased boiling point

CLQ244

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7. Identify the mixture that shows positive deviations from Raoult's Law **[JEE (Main) 2024]**

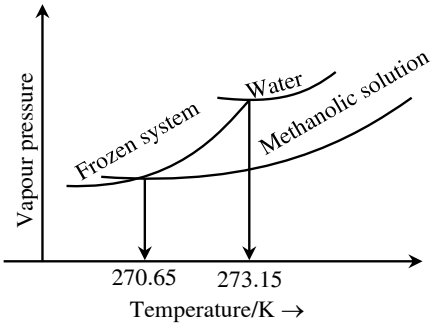
(1) $(CH_3)_2CO + C_6H_5NH_2$
(2) $CHCl_3 + C_6H_6$
(3) $CHCl_3 + (CH_3)_2CO$
(4) $(CH_3)_2CO + CS_2$

CLQ245

8. Mass of ethylene glycol (antifreeze) to be added to 18.6 kg of water to protect the freezing point at –24°C is_____ kg (Molar mass in g mol^{-1} for ethylene glycol 62, K_f of water = 1.86 K kg mol^{-1}) **[JEE (Main) 2024]**

CLQ259

9. When 'x' × 10^{-2} mL methanol (molar mass = 32 g; density = 0.792 g/cm^3) is added to 100 mL water (density = 1 g/cm^3), the following diagram is obtained.



x =.....(nearest integer)
[Given: Molal freezing point depression constant of water at 273.15 K is 1.86 K kg mol^{-1}] **[JEE (Main) 2024]**

CLQ246

For More Join: @IITJEE_Advanced

10. The vapour pressure of pure benzene and methyl benzene at 27°C is given as 80 Torr and 24 Torr, respectively. The mole fraction of methyl benzene in vapour phase, in equilibrium with an equimolar mixture of those two liquids (ideal solution) at the same temperature is___× 10^{-2} (nearest integer)

[JEE (Main) 2024]

CLQ247

11. 2.5 g of a non-volatile, non-electrolyte is dissolved in 100 g of water at 25°C. The solution showed a boiling point elevation by 2°C. Assuming the solute concentration in negligible with respect to the solvent concentration, the vapour pressure of the resulting aqueous solution is _____ mm of Hg (nearest integer)

[Given : Molal boiling point elevation constant of water (K_b) = 0.52 K. kg mol^{-1},
1 atm pressure = 760 mm of Hg, molar mass of water = 18 g mol^{-1}]

[JEE (Main) 2024]

CLQ248

12. 2.7 Kg of each of water and acetic acid are mixed, The freezing point of the solution will be –x °C. Consider the acetic acid does not dimerise in water, nor dissociates in water x = _____(nearest integer)

[Given : Molar mass of water = 18 g mol^{-1}, acetic acid = 60 g mol^{-1}]

^{K_f} H_2O : 1.86 K kg mol^{-1}

^{K_f} acetic acid : 3.90 K kg mol^{-1}

freezing point : H_2O = 273 K, acetic acid = 290 K] **[JEE (Main) 2024]**

CLQ249

Rankers Academy JEE

13. The osmotic pressure of a dilute solution is 7 × 10^5 Pa at 273K. Osmotic pressure of the same solution at 283K is___× 10^4 Nm^{-2}.

[JEE (Main) 2024]

CLQ251

14. What happens to freezing point of benzene when small quantity of napthalene is added to benzene? **[JEE (Main) 2024]**

(1) Increases (2) Remains unchanged

(3) First decreases and then increases (4) Decreases

CLQ252

15. Sea water contains 29.25% NaCl and 19% $MgCl_2$ by weight of solution. The normal boiling point of the sea water is ____°C (Nearest integer)

Assume 100% ionization for both NaCl and $MgCl_2$

Given : $K_b(H_2O)$ = 0.52 K kg mol^{-1}

Molar mass of NaCl and $MgCl_2$ is 58.5 and 95 g mol^{-1} respectively. **[JEE (Main) 2023]**

CLQ135

16. What weight of glucose must be dissolved in 100 g of water to lower the vapour pressure by 0.20 mm Hg? (Assume dilute solution is being formed)

Given : Vapour pressure of pure water is 54.2 mm Hg at room temperature. Molar mass of glucose is 180 g mol^{-1}. **[JEE (Main) 2023]**

(1) 4.69 g (2) 3.59 g (3) 2.59 g (4) 3.69 g

CLQ136

For More Join: @IITJEE_Advanced

17. In the depression of freezing point experiment

A. Vapour pressure of the solution is less than that of pure solvent

B. Vapour pressure of the solution is more than that of pure solvent

C. Only solute molecules solidify at the freezing point

D. Only solvent molecules solidify at the freezing point **[JEE (Main) 2023]**

(1) A and D only (2) B and C only (3) A and C only (4) A only

CLQ049

18. 1 kg of 0.75 molal aqueous solution of sucrose can be cooled up to –4°C before freezing. The amount of ice (in g) that will be separated out is _____. (Nearest integer)

[Given : $K_f(H_2O) = 1.86$ K kg mol^{-1}] **[JEE (Main) 2021]**

CLQ046

19. If 250 cm^3 of an aqueous solution containing 0.73 g of a protein A is isotonic with one litre of another aqueous solution containing 1.65 g of a protein B, at 298 K, the ratio of the molecular mases of A and B is _____ × 10^{-2} (to the nearest integer). **[JEE (Main) 2020]**

CLQ043

20. Elevation in the boiling point for 1 molal solution of glucose is 2 K. The depression in the freezing point of 2 molal solutions of glucose in the same solvent is 2 K. The relation between K_b and K_f is: **[JEE (Main) 2019]**

(1) $K_b = 0.5\ K_f$ (2) $K_b = 2\ K_f$ (3) $K_b = 1.5\ K_f$ (4) $K_b = K_f$

CLQ033

21. 1 g of non-volatile non-electrolyte solute is dissolved in 100g of two different solvents A and B whose ebullioscopic constants are in the ratio of 1 : 5. The ratio of the elevation in their boiling points, $\frac{\Delta T_b(A)}{\Delta T_b(B)}$, is : **[JEE (Main) 2019]**

(1) 5 : 1 (2) 10 : 1 (3) 1 : 5 (4) 1 : 0.2

CLQ038

22. When a non-volatile solute is added to the solvent, the vapour pressure of the solvent decreases by 10 mm of Hg. The mole fraction of the solute in the solution is 0.2. What would be the mole fraction of the solvent if decrease in vapour pressure is 20 mm of Hg ? **[JEE (Main) 2025]**

(1) 0.6 (2) 0.4 (3) 0.2 (4) 0.8

CLQ261

23. Considering acetic acid dissociates in water, its dissociation constant is 6.25 × 10^{-5}. If 5 mL of acetic acid is dissolved in 1 litre water, the solution will freeze at –x × 10^{-2} °C, provided pure water freezes at 0 °C.

x = _____. (Nearest integer)

Given : $(K_f)_{water} = 1.86$ K kg mol^{-1}.

density of acetic acid is 1.2 g mol^{-1}

molar mass of water = 18 g mol^{-1}.

molar mass of acetic acid = 60 g mol^{-1}.

density of water = 1 g cm^{-3}

Acetic acid dissociates as

$$CH_3COOH \rightleftharpoons CH_3COO^{\ominus} + H^{\oplus}$$

[JEE (Main) 2024]

CLQ253

For More Join: @IITJEE_Advanced

24. Consider the dissociation of the weak acid HX as given below

$HX(aq) \rightleftharpoons H^+(aq) + X^-(aq)$, $K_a = 1.2 \times 10^{-5}$

[K_a : dissociation constant]

The osmotic pressure of 0.03 M aqueous solution of HX at 300 K is ____ $\times 10^{-2}$ bar (nearest integer).

[Given : R = 0.083 L bar Mol^{-1} K^{-1}]

[JEE (Main) 2024]

CLQ254

25. A solution containing 10g of an electrolyte AB_2 in 100g of water boils at 100.52°C. The degree of ionization of the electrolyte (α) is ____ $\times 10^{-1}$.

(nearest integer)

[Given : Molar mass of AB_2 = 200g mol^{-1}. K_b (molal boiling point elevation const. of water) = 0.52 K kg mol^{-1}, boiling point of water = 100°C ; AB_2 ionises as $AB_2 \rightarrow A^{2+} + 2B^-$]

[JEE (Main) 2024]

CLQ255

26. 0.05M $CuSO_4$ when treated with 0.01M $K_2Cr_2O_7$ gives green colour solution of $Cu_2Cr_2O_7$.

The [SPM : Semi Permeable Membrane]

$K_2Cr_2O_7$	$CuSO_4$
Side X	SPM Side Y

Due to osmosis :

[JEE (Main) 2024]

(1) Green colour formation observed on side Y.

(2) Green colour formation observed on side X.

(3) Molarity of $K_2Cr_2O_7$ solution is lowered.

(4) Molarity of $CuSO_4$ solution is lowered.

CLQ256

27. The solution from the following with highest depression in freezing point/lowest freezing point is

[JEE (Main) 2024]

(1) 180 g of acetic acid dissolved in water

(2) 180 g of acetic acid dissolved in benzene

(3) 180 g of benzoic acid dissolved in benzene

(4) 180 g of glucose dissolved in water

CLQ258

28. 1.22 g of an organic acid is separately dissolved in 100 g of benzene (K_b = 2.6 K kg mol^{-1}) and 100 g of acetone (K_b = 1.7 K kg mol^{-1}). The acid is known to dimerize in benzene but remain as a monomer in acetone. The boiling point of the solution in acetone increases by 0.17°C.

The increase in boiling point of solution in benzene in °C is x $\times 10^{-2}$. The value of x is _____.

(Nearest integer)

[Atomic mass : C = 12.0, H = 1.0, O = 16.0]

[JEE (Main) 2021]

CLQ047

For More Join: @IITJEE_Advanced

29. How much amount of NaCl should be added to 600 g of water (ρ = 1.00 g/mL) to decrease the freezing point of water to – 0.2 °C ? _____. (The freezing point depression constant for water = 2K kg mol^{-1}) **[JEE (Main) 2020]**

CLQ042

30. The osmotic pressure of a solution of NaCl is 0.10 atm and that of a glucose solution is 0.20 atm. The osmotic pressure of a solution formed by mixing 1 L of the sodium chloride solution with 2 L of the glucose solution is $x \times 10^{-3}$ atm. x is ____. (nearest integer) : **[JEE (Main) 2020]**

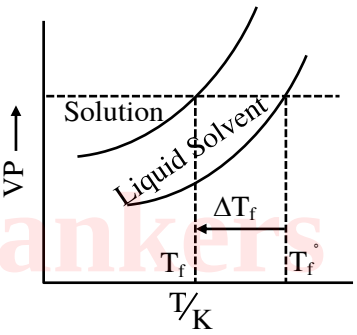
CLQ044

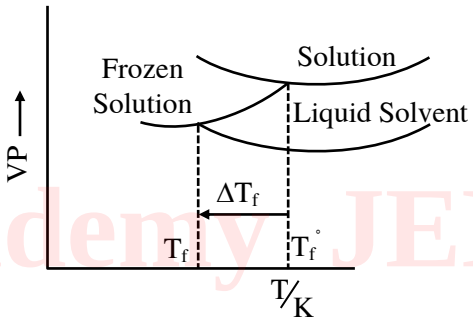
31. The osmotic pressure of a dilute solution of an ionic compound XY in water is four times that of a solution of 0.01 M $BaCl_2$ in water. Assuming complete dissociation of the given ionic compounds in water, the concentration of XY (in mol L^{-1}) in solution is : **[JEE (Main) 2019]**

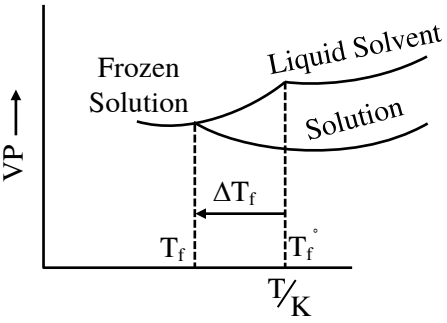
(1) 6×10^{-2} (2) 4×10^{-4} (3) 16×10^{-4} (4) 4×10^{-2}

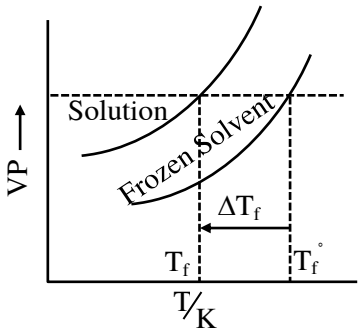
CLQ036

32. Consider the given plots of vapour pressure (VP) vs temperature (T/K) Which amongst the following options is correct graphical representation showing ΔT_f, depression in the freezing point of solvent in a solution ? **[JEE (Main) 2025]**

(1)


(2)


(3)


(4)


CLQ262

33. What is the freezing point depression constant of a solvent, 50 g of which contain 1 g non volatile solute (molar mass 256 g mol^{-1}) and the decrease in freezing point is 0.40 K? **[JEE (Main) 2025]**

(1) 5.12 K kg mol^{-1} (2) 4.43 K kg mol^{-1} (3) 1.86 K kg mol^{-1} (4) 3.72 K kg mol^{-1}

CLQ263

34. Arrange the following solutions in order of their increasing boiling points. **[JEE (Main) 2025]**

(i) 10^{-4} M NaCl (ii) 10^{-4} M Urea (iii) 10^{-3} M NaCl (iv) 10^{-2} M NaCl

(1) (ii) < (i) < (iii) < (iv) (2) (ii) < (i) $\cong$ (iii) < (iv)

(3) (i) < (ii) < (iii) < (iv) (4) (iv) < (iii) < (i) < (ii)

CLQ264

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For More Join: @IITJEE_Advanced

35. $CrCl_3.xNH_3$ can exist as a complex. 0.1 molal aqueous solution of this complex shows a depression in freezing point of 0.558ºC. Assuming 100% ionisation of this complex and coordination number of Cr is 6, the complex will be
(Given K_f = 1.86 K kg mol^{-1}) **[JEE (Main) 2025]**
(1) $[Cr(NH_3)_6]Cl_3$ (2) $[Cr(NH_3)_4Cl_2]Cl$ (3) $[Cr(NH_3)_5Cl]Cl_2$ (4) $[Cr(NH_3)_3Cl_3]$

CLQ265

36. The observed and normal masses of compound MX_2 are 65.6 and 164 respectively. The percent degree of ionisation of MX_2 is _____%. (Nearest integer) **[JEE (Main) 2025]**

CLQ266

37. 1.24 g of AX_2 (molar mass 124 g mol^{-1}) is dissolved in 1 kg of water to form a solution with boiling point of 100.0156°C, while 25.4 g of AY_2 (molar mass 250 g mol^{-1}) in 2 kg of water constitutes a solution with a boiling point of 100.0260°C.
$K_b(H_2O)$ = 0.52 K kg mol^{-1}
Which of the following is **correct** ? **[JEE (Main) 2025]**
(1) AX_2 and AY_2 (both) are completely unionised.
(2) AX_2 and AY_2 (both) are fully ionised.
(3) AX_2 is completely unionised while AY_2 is fully ionised.
(4) AX_2 is fully ionised while AY_2 is completely unionised.

CLQ267

38. If A_2B is 30% ionised in an aqueous solution, then the value of van't Hoff factor (i) is ____$\times 10^{-1}$.
[JEE (Main) 2025]

CLQ268

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For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

1. Liquids **A** and **B** form ideal solution for all compositions of **A** and **B** at 25°C. Two such solutions with 0.25 and 0.50 mole fractions of **A** have the total vapor pressures of 0.3 and 0.4 bar, respectively. What is the vapor pressure of pure liquid **B** in bar? **[JEE (Advanced) 2020]**

CLQ081

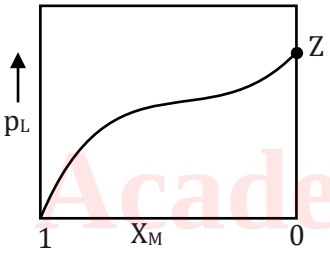
2. Liquids A and B form ideal solution over the entire range of composition. At temperature T, equimolar binary solution of liquids A and B has vapour pressure 45 Torr. At the same temperature, a new solution of A and B having mole fractions X_A and X_B, respectively, has vapour pressure of 22.5 Torr. The value of X_A/X_B in the new solution is _____.
(Given that the vapour pressure of pure liquid A is 20 Torr at temperature T)

[JEE (Advanced) 2018]

CLQ078

3. For a solution formed by mixing liquids L and M, the vapour pressure of L plotted against the mole fraction of M in solution is shown in the following figure, here X_L and X_M represent mole fractions of L and M, respectively, in the solution. the correct statement(s) applicable to this system is(are)

[JEE (Advanced) 2017]



Rankers Academy JEE

(A) Attractive intramolecular interactions between L–L in pure liquid L and M–M in pure liquid M are stronger than those between L–M when mixed in solution

(B) The point Z represents vapour pressure of pure liquid M and Raoult's law is obeyed when $X_L \to 0$

(C) The point Z represents vapour pressure of pure liquid L and Raoult's law is obeyed when $X_L \to 1$

(D) The point Z represents vapour pressure of pure liquid M and Raoult's law is obeyed from $X_L = 0$ to $X_L = 1$

CLQ076

4. Benzene and naphthalene form an ideal solution at room temperature. For this process, the true statement(s) is(are) **[JEE (Advanced) 2013]**

(A) ΔG is positive

(B) ΔS_{system} is positive

(C) $\Delta S_{surroundings} = 0$

(D) $\Delta H = 0$

CLQ073

5. Mixture(s) showing positive deviation from Raoult's law at 35°C is (are)

[JEE (Advanced) 2016]

(A) carbon tetrachloride + methanol

(B) carbon disulphide + acetone

(C) benzene + toluene

(D) phenol + aniline

CLQ075

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6. 50 mL of 0.2 molal urea solution (density = 1.012 g mL^{-1} at 300 K) is mixed with 250 mL of a solution containing 0.06 g of urea. Both the solutions were prepared in the same solvent. The osmotic pressure (in Torr) of the resulting solution at 300 K is _____.
[Use : Molar mass of urea = 60 g mol^{-1}; gas constant, R = 62 L Torr K^{-1} mol^{-1} ;
Assume, $\Delta_{mix}H = 0$, $\Delta_{mix}V = 0$] **[JEE (Advanced) 2023]**

CLQ137

Question Nos. 7 and 8

The boiling point of water in a 0.1 molal silver nitrate solution (solution **A**) is **x**°C. To this solution **A**, an equal volume of 0.1 molal aqueous barium chloride solution is added to make a new solution **B**. The difference in the boiling points of water in the two solutions **A** and **B** is **y** × 10^{-2} °C.
(Assume : Densities of the solutions **A** and **B** are the same as that of water and the soluble salts dissociate completely.)
Use: Molal elevation constant (Ebullioscopic Constant), K_b = 0.5 K kg mol^{-1}; Boiling point of pure water as 100°C.)

7. The value of **x** is _____. **[JEE (Advanced) 2021]**

CLQ082

8. The value of $|\mathbf{y}|$ is ____ . **[JEE (Advanced) 2021]**

CLQ083

9. On dissolving 0.5 g of a non-volatile non-ionic solute to 39 g of benzene, its vapor pressure decreases from 650 mm Hg to 640 mm Hg. The depression of freezing point of benzene (in K) upon addition of the solute is ____ .
(Given data : Molar mass and the molal freezing point depression constant of benzene are 78 g mol^{-1} and 5.12 K kg mol^{-1}, respectively) **[JEE (Advanced) 2019]**

CLQ080

10. For a dilute solution containing 2.5 g of a non-volatile non-electrolyte solute in 100 g of water, the elevation in boiling point at 1 atm pressure is 2°C. Assuming concentration of solute is much lower than the concentration of solvent, the vapour pressure (mm of Hg) of the solution is
(take K_b=0.76 K kg mol^{-1}) **[JEE (Advanced) 2012]**
(A) 724 (B) 740 (C) 736 (D) 718

CLQ072

11. Vessel-1 contains $\mathbf{w_2}$ g of a non-volatile solute **X** dissolved in $\mathbf{w_1}$ g of water. Vessel-2 contains $\mathbf{w_2}$ g of another non-volatile solute **Y** dissolved in $\mathbf{w_1}$ g of water. Both the vessels are at the same temperature and pressure. The molar mass of **X** is 80% of that of **Y**. The van't Hoff factor for **X** is 1.2 times of that of **Y** for their respective concentrations.
The elevation of boiling point for solution in Vessel-1 is _____% of the solution in Vessel-2.
[JEE (Advanced) 2024]

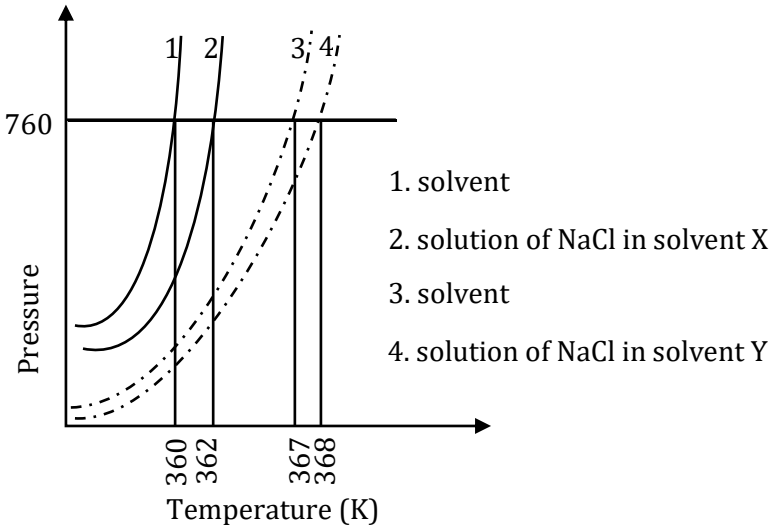
CLQ260

12. An aqueous solution is prepared by dissolving 0.1 mol of an ionic salt in 1.8 kg of water at 35°C. The salt remains 90% dissociated in the solution. The vapour pressure of the solution is 59.724 mm of Hg. Vapor pressure of water at 35°C is 60.000 mm of Hg. The number of ions present per formula unit of the ionic salt is _____. **[JEE (Advanced) 2022]**

CLQ084

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13. The plot given below shows P–T curves (where P is the pressure and T is the temperature) for two solvents X and Y and isomolal solutions of NaCl in these solvents. NaCl completely dissociates in both the solvents.



On addition of equal number of moles a non-volatile solute S in equal amount (in kg) of these solvents, the elevation of boiling point of solvent X is three times that of solvent Y. Solute S is known to undergo dimerization in these solvents. If the degree of dimerization is 0.7 in solvent Y, the degree of dimerization in solvent X is _____. **[JEE (Advanced) 2018]**

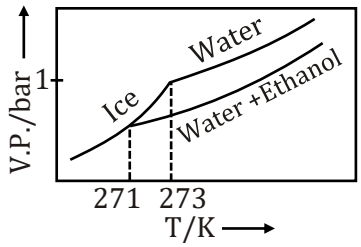
CLQ079

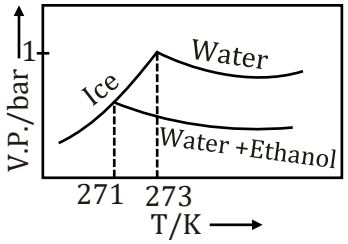
Rankers Academy JEE

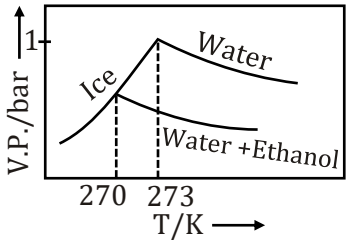
14. Pure water freezes at 273 K and 1 bar. The addition of 34.5 g of ethanol to 500 g of water changes the freezing point of the solution. Use the freezing point depression constant of water as 2 K kg mol^{-1}. The figures shown below represents plots of vapour pressure (V.P.) versus temperature (T). [Molecular weight of ethanol is 46 g mol^{-1}]

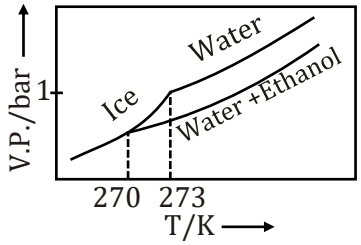
Among the following, the option representing change in the freezing point is –

[JEE (Advanced) 2017]

(A)


(B)


(C)


(D)


CLQ077

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15. If the freezing point of a 0.01 molal aqueous solution of a cobalt (III) chloride-ammonia complex (which behaves as a strong electrolyte) is – 0.0558°C, the number of chloride (s) in the coordination sphere of the complex is- [K_f of water = 1.86 K kg mol^{-1}] **[JEE (Advanced) 2015]**

CLQ074

16. The freezing point (in °C) of a solution containing 0.1 g of $K_3[Fe(CN)_6]$ (Mol. Wt. 329) in 100 g of water (K_f = 1.86 K kg mol^{-1}) is - **[JEE (Advanced) 2011]**

(A) -2.3×10^{-2} (B) -5.7×10^{-2} (C) -5.7×10^{-3} (D) -1.2×10^{-2}

CLQ071

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	C	A	D	A	A	C	B	A	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	C	A	B	C	C	C	C	B	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	D	D	C	A	A	C	C	A	A
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	C	A	A	A	B	C	D	A	D	D
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	C	B	D	B	D	D	B	B	B	A

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9
Ans.	CD	ABCD	ABC	B	C	D	C	A	C
Que.	10								
Ans.	(A) → P ; (B) → Q, R, S ; (C) → T ; (D) → P, Q, R, S, T								

EXERCISE (S)

1.	3	**2.**	64	**3.**	777	**4.**	150	**5.**	2048
6.	40	**7.**	78	**8.**	60	**9.**	0.04	**10.**	1.50
11.	0.25	**12.**	0.25	**13.**	0.50	**14.**	6.00	**15.**	0.73
16.	5.00	**17.**	0.04	**18.**	0.32	**19.**	17.38	**20.**	5.10
21.	38.71	**22.**	49.26	**23.**	3.69	**24.**	$(4.8 \times 10^5$ g/mol)		
25.	–3.53	**26.**	0.125	**27.**	$(7.482 \times 10^5$ Nm$^{-2})$				
28.	2.00	**29.**	0.50	**30.**	12.00				

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	3	3	600	3	4	4	15	543	23
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	707	31	(72.56) or (73)	4	116	4	1	518	177	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	3	1	19	76	5	4	1	13	(1.74 - 1.76)	167
Que.	31	32	33	34	35	36	37	38		
Ans.	1	3	1	1	3	75	4	16		

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	0.20	19	AC	BCD	AB	682	100.10	2.50	1.02 or 1.03	A
Que.	11	12	13	14	15	16				
Ans.	150	5	0.05	D	1	A				

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